A Bedside Decision Support System for Disorders of Consciousness: Multi-Scale Temporal Dynamics and Multicenter Validation

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***Abstract*—Bedside EEG summaries often discard temporal order, limiting assessment of residual consciousness in small clinical cohorts. We developed an interpretable framework combining physically adjacent lag-1 persistence (ρ_1), normalized jump rate (D_1), and common-mode-adjusted bilateral correlation (r_{residual}) with fixed-weight consensus scoring. Evaluation comprised two continuous EEG cohorts, one exploratory short-record dataset, and one sleep cohort ($N = 129$ records). On long-term discovery EEG ($N = 41$), the primary four-feature suite achieved 0.9324 AUROC under 25 subject-level outer partitions, compared with 0.8964 for the strongest supervised baseline using identical features and preprocessing (paired gain $+0.0360$, 95% interval $[+0.0090, +0.0728]$). Sequence dependence of the lag-1 extension was verified by 10,000 synchronous shuffles, reducing its AUROC from 0.8911 to 0.8182 ± 0.0049 while preserving channel marginals and instantaneous spatial covariance. On China Rehabilitation Research Center telemetry ($N = 15$), shared unlabeled target standardization and direction alignment yielded 0.7778 AUROC; fixed soft truncation ($c = 1.5$) increased this to 0.7963, with matched supervised baselines at 0.7407–0.7593. On 120-s Figshare records ($N = 50$), the jump-rate branch achieved 0.7691 AUROC, comparable to matched baselines at 0.7674–0.7759. Sleep-stage transition flexibility in the Wielek cohort ($N = 23$) provided complementary evidence across circadian timescales. With 2.15 ms per evaluated update in cached-power replay, the framework provides a compact, auditable basis for bedside assessment and prioritization of further multimodal evaluation.**

***Index Terms*—Disorders of consciousness (DoC), electroencephalography (EEG), temporal dynamics, clinical decision support systems (CDSS), mesocircuit hypothesis, consensus fusion.**

I. Introduction

SEVERE acute brain injury from trauma, anoxia, or stroke leaves patients in prolonged disorders of consciousness (DoC), clinically categorized into the vegetative state/unresponsive wakefulness syndrome (VS/UWS) and the minimally conscious state (MCS) [1], [2]. Distinguishing MCS from VS/UWS informs decisions on withdrawing life-sustaining treatment, administering analgesia, and projecting rehabilitation potential. Bedside clinical evaluations carry an estimated 40% misdiagnosis rate [3]. Although standardized assessments such as the Coma Recovery Scale-Revised (CRS-R) [3] reduce diagnostic error, motor impairment and fluctuating arousal frequently confound behavioral examinations [4]. Furthermore, prospective cohorts establish that 15%–35% of behaviorally unresponsive patients exhibit cognitive-motor dissociation (CMD) [5], retaining volitional cortical responses despite absent motor output.

Functional neuroimaging and perturbational protocols (fMRI, ^{18}F-FDG PET, TMS-EEG) [6], [7] show that preserved cortical complexity differentiates MCS from VS/UWS, but the hazards of intra-hospital transport preclude routine bedside deployment in the intensive care unit. Sparse continuous EEG (4–8 electrodes) provides the primary practical modality for longitudinal monitoring [8]–[10]. Beyond static power spectra, prior investigations examined non-linear entropy [8], [11], microstates [12], sleep architecture [13], [14], long-range temporal correlations via detrended fluctuation analysis (DFA) [15], [16], source-space connectivity [17], and dynamical criticality [18]. However, clinical bedside translation remains constrained: multi-scale DFA requires stationary dense montages; source reconstruction is ill-posed with sparse arrays; entropy measures exhibit marked artifact sensitivity; and flexible classifiers risk overfitting under small discovery cohorts ($N_{\text{train}} \approx 32$). Bedside assessment has therefore relied on empirical band power, marginal dispersion ($-\text{CV}_\theta$), and unadjusted bilateral correlations [10], [13]. As a result, non-invasive bedside discrimination has remained bounded to a moderate plateau (0.75–0.82 AUROC).

A. The Permutation Invariance Defect of Conventional Metrics

This diagnostic plateau stems directly from a formal mathematical property shared by standard bedside metrics: strict permutation invariance. Conventional bedside pipelines characterize cortical dynamics primarily through aggregate dispersion, such as theta power variation ($-\text{CV}_\theta$), and instantaneous bilateral cross-correlation (r_{interhem}) evaluated over multi-second epochs. Both sample variance and Pearson correlation are mathematically invariant under arbitrary permutations of the temporal index. Randomly shuffling the chronological sequence of EEG epochs alters neither metric, preserving their numerical values identically within machine precision ($\approx 10^{-15}$). Consequently, these formulations treat continuous electrophysiological trajectories as unordered sets of observations, discarding multi-second sequence continuity and directional temporal reverberation.

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B. The Mesocircuit Hypothesis and Autoregressive Memory

This methodological gap diverges from established neurobiology of conscious recovery, specifically the anterior forebrain mesocircuit hypothesis [19]. Under this framework, sustained behavioral awareness depends on recurrent reciprocal excitatory loops linking the central thalamus, striatum, and frontoparietal association cortices. Preserved mesocircuit connectivity manifests electrophysiologically as multi-second autoregressive continuity, quantifiable via lag-1 autocorrelation (ρ_1) across consecutive epochs. In VS/UWS, widespread structural deafferentation and loss of striatal inhibition disinhibit the internal globus pallidus (GPi), which drives sustained hyperpolarization of the central thalamus and degrades cortical activity into memoryless fluctuations. Differentiating conscious capacity from reflexive unresponsiveness therefore requires operators sensitive to physical sequence persistence.

As illustrated in Fig. 1a, individual theta trajectories in MCS (e.g., patient P36) exhibit sustained multi-second wave-holding, whereas VS/UWS trajectories (e.g., patient P25) break into decoupled transients. Static dispersion ($-\mathrm{CV}_\theta$) displays substantial individual overlap between cohorts ($p = 0.0264$; Fig. 1c), whereas lag-1 persistence (ρ_1) reliably separates conscious from unresponsive states ($p = 0.0227$; Fig. 1b).

While continuous multi-hour telemetry enables stable asymptotic estimation of ρ_1, emergency bedside triage frequently relies on ultra-short recordings (< 5 min). In this small-sample regime, sample autocorrelation exhibits elevated estimator variance ($\mathrm{Var}(\hat{\rho}_1) \sim \mathcal{O}(1/M)$). For brief recordings, we introduce the normalized jump rate (D_1) across verified adjacent epochs ($\mathbb{I}_{\text{adj}} = 1$), which analytically unifies static amplitude dispersion and temporal decay via $D_1 \approx 2\pi^{-1/2}(-\mathrm{CV}_\theta)(1-\rho_1)^{1/2}$.

C. Spurious Bilateral Coupling Driven by Diffuse Slow Waves

A second confound in bedside DoC monitoring is diffuse slow-wave volume conduction. In severe acute brain injury, high-amplitude delta and theta oscillations propagate broadly across the scalp via volume conduction and synchronized subcortical driving. Although spatial filters such as the common average reference (CAR) subtract instantaneous linear potential offsets, rectified band-power envelopes retain substantial shared variance from global pacemaking ($R^2 \approx 47\%$). This background inflates raw inter-hemispheric correlation and obscures regional differentiation. Decoupling this shared envelope via ordinary least squares (OLS) orthogonal projection isolates residual bilateral resonance (r_{residual}), separating widespread pathological co-activation in VS/UWS from preserved modular dynamics in MCS.

D. Contributions of This Work

This study introduces an interpretable bedside decision support framework with three principal contributions:

1) *Order-aware temporal operators and artifact decoupling*: We formulate lag-1 autocorrelation (ρ_1) for continuous telemetry alongside normalized adjacent jump rates (D_1) for short recordings under physical adjacency constraints ($\mathbb{I}_{\text{adj}} = 1$), analytically unified via $D_1 \approx 2\pi^{-1/2}(-\mathrm{CV}_\theta)(1-\rho_1)^{1/2}$. Synchronously shuffling 10,000 surrogate sequences reduces classification performance from 0.8911 AUROC to permutation baseline (0.8182 ± 0.0049, $p < 0.0001$), confirming that diagnostic separation depends on physical sequence order rather than static marginal distributions. Ordinary least squares (OLS) orthogonal projection decouples shared slow-wave envelope variance ($R^2 \approx 47\%$) to isolate genuine bilateral resonance (r_{residual}).

2) *Interpretable, parameter-free bedside consensus*: The parsimonious four-feature suite $\mathbf{x}_{\text{ortho}} = [c_{\text{global}}, -\mathrm{CV}_\theta, \rho_1, r_{\text{residual}}]^\top$ attains 0.9324 AUROC and 0.8998 average precision (AP) on primary discovery telemetry. Given identical feature inputs, equal-weight consensus scoring exceeds the top-performing matched supervised model (logistic regression) by $+0.0360$ AUROC (Dirichlet paired 95% confidence interval $[+0.0090, +0.0728]$), requiring 2.15 ms per evaluated update in cached-power replay (Supplementary Note D).

3) *Multicenter evaluation across distinct acquisition settings*: The framework is validated across 129 records spanning four independent cohorts: CRRC telemetry ($N = 15$) establishes cross-hardware generalization (0.7778 to 0.7963 AUROC); Figshare 120-second recordings ($N = 50$) demonstrate emergency triage utility (0.7691 AUROC); and Wielek polysomnography ($N = 23$) confirms preservation of macro-scale sleep stage transitions across circadian cycles.

II. Mathematical Formulation and Methods

Fig. 2 summarizes the processing, scoring and evaluation workflow.

A. Clinical Cohorts and Ethical Compliance

Four multicenter cohorts totaling $N = 129$ records were evaluated under institutional ethical approvals:

1) Mendeley Discovery Cohort ($N = 41$): Continuous 6-channel bedside EEG (F3, F4, C3, C4, O1, O2; 256 Hz) from 41 patients with severe brain injuries at FRCCICMR Moscow [20]: 12 MCS (8 MCS$^+$, 4 MCS$^-$; duration 6.00 ± 0.00 h) and 29 VS/UWS (5.97 ± 0.61 h), diagnosed through repeated CRS-R assessments.

2) CRRC Inpatient Cohort ($N = 15$): Prospective bedside telemetry from 15 subacute male patients (6 MCS, 9 VS/UWS; 11 ICH, 4 TBI; age 51.7 ± 12.3 years, range 28–69; recording duration 19.5 ± 8.0 min; Ethics No. 2025-118-01 [21]) at China Rehabilitation Research Center (CRRC / Beijing Bo'ai Hospital). Data were acquired with a 32-channel NeuSen NSM2 system (Neuracle, 1000 Hz, 4-second blocks [4,000 samples]), and CRS-R was confirmed by consensus. As detailed in the supplementary material, recordings were acquired during stable post-acute rehabilitation; daily medication logs were unavailable. Homologous 6 channels were evaluated; in 2 patients (CRRC_008 and CRRC_009), mechanical drift at O_2 corrupted bilateral derivation, leaving a pristine 6-channel subgroup of $N = 13$ (6 MCS including 1 EMCS, 7 VS/UWS).

3) Figshare Short-Record Analysis ($N = 50$): High-density resting-state EEG (250 Hz, 120-s) from 50 patients with prolonged DoC (19 MCS, 31 VS/UWS) at Hangzhou Mingzhou

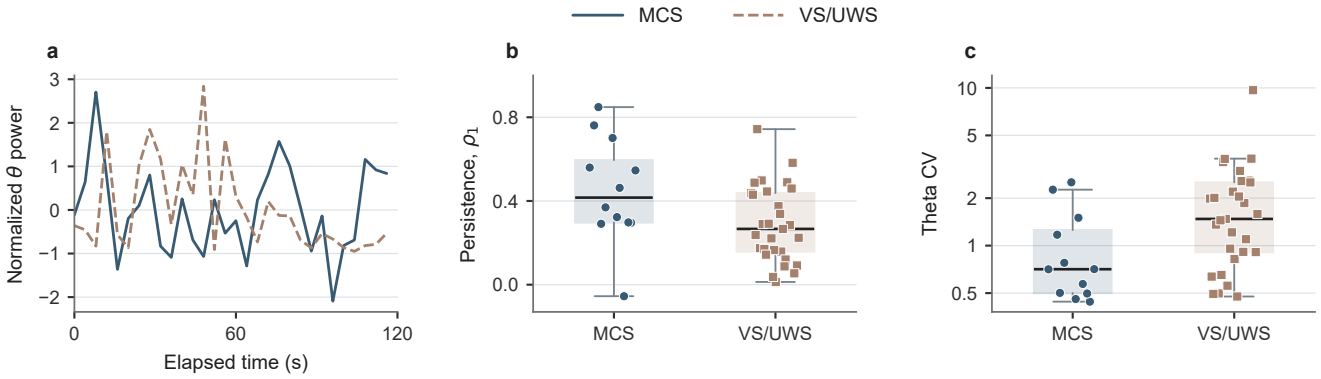


Fig. 1. Bedside EEG dynamics and patient distributions. (a) Theta power in 30 consecutive 4-s blocks from P36 (MCS) and P25 (VS/UWS), standardized within each trace. (b,c) Lag-1 persistence and positive theta CV for 12 MCS and 29 VS/UWS patients. Jittered points include all patients and outliers. Boxes show medians and interquartile ranges (IQR); whiskers reach the most extreme values within 1.5 IQR. Panel c uses a logarithmic axis and displays the sign reversal of $-\mathrm{CV}_\theta$. Two-sided Mann–Whitney tests give unadjusted $p = 0.0227$ and $p = 0.0264$, respectively.

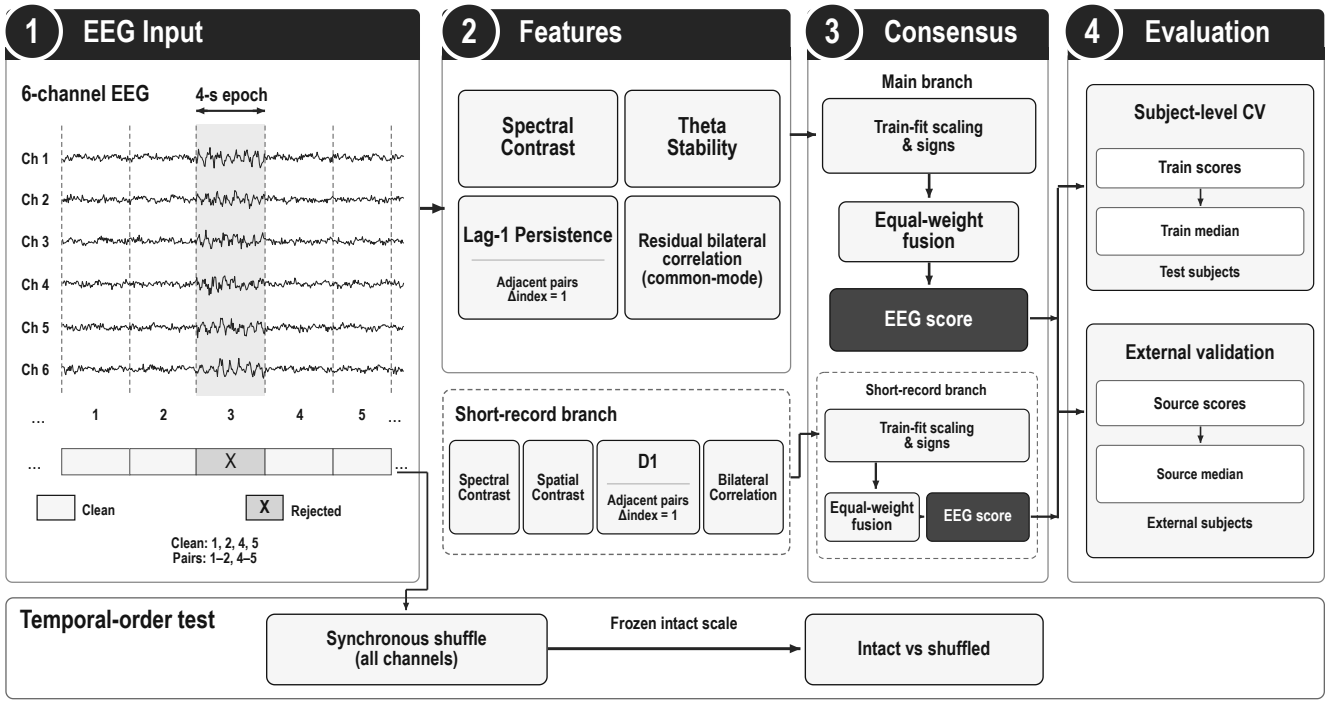


Fig. 2. System overview. The EEG traces are schematic. Original block indices are retained for adjacent-pair features. The main and short-record branches use separate four-feature suites and equal-weight scores. Discovery uses 4-s epochs; the Figshare analysis uses 2-s segments. Source training data determine scaling, signs and median thresholds. External protocols additionally include target-batch standardization and specified direction alignment where stated. The temporal-order test uses synchronous multichannel shuffles and intact-fitted scales for the five-feature lag-1 extension.

Brain Rehabilitation Hospital [12], [22], evaluating frozen-source scoring on homologous channels. Published group counts support the 19 MCS and 31 VS/UWS composition; file-level diagnosis mapping and intersegment continuity require independent confirmation.

4) Wielek PSG Cohort ($N = 23$): 24-hour polysomnography from 23 post-comatose patients (11 MCS, 12 VS/UWS) [13], with 30-second hypnograms (Wake, N1, N2, N3, REM) tracking macro-scale transition continuity across circadian sleep cycles.

B. EEG Acquisition, Preprocessing, and Physical Adjacency Constraint

Continuous 6-channel EEG (256 Hz) was re-referenced to Common Average Reference (CAR) and segmented into non-overlapping 4-second epochs (1,024 samples). Spectral densities were computed across each epoch via Hann-tapered periodograms ($\Delta f = 0.25$ Hz), extracting theta (4–8 Hz) and alpha (8–13 Hz) band power.

Primary artifact screening at $500\,\mu$V retained $5{,}185.5 \pm 595.6$ clean blocks per patient ($96.98\% \pm 3.2\%$, 5.76 ± 0.66 h; rejection rate $2.84\% \pm 2.1\%$ in MCS vs. $3.09\% \pm 3.6\%$ in VS/UWS, $p = 0.82$, Mann-Whitney U test). Tightening the

TABLE I
PERFORMANCE OF ORDER-AWARE TEMPORAL DYNAMICS, CONSENSUS FUSION, AND MACHINE LEARNING BASELINES ON MENDELEY COHORT (25 OUTER PARTITIONS; 95% CIS PAIRED VS. CANONICAL LINEAR BASELINE VIA 10,000 DIRICHLET RESAMPLES; PR-AUC REPORTED AS AVERAGE PRECISION [AP], WITH TRAPEZOIDAL PR-AUC = 0.8782 FOR 4-FEAT LINEAR; MATCHED SUPERVISED ROWS USE THE SAME FOUR FEATURES AS THE DEPLOYMENT SUITE; 5-FEAT COMBINED MODELS SUBSTITUTE r_{INTERHEM} WITH r_{RES}).

Model / Feature Configuration	AUROC	PR-AUC	Balanced Acc.	Sensitivity	Specificity	95% CI vs. Baseline
Consensus Aggregation Frameworks						
Canonical Static (4 Feats, Linear)	0.8200	0.7823	0.7201	0.8333	0.6069	Baseline
Canonical Static (4 Feats, Rank Consensus)	0.7893	0.7299	0.7270	0.8333	0.6207	$[-0.0619, -0.0076]$
Plus Lag-1 ρ_1 (5 Feats, Linear)	**0.8911**	**0.8384**	**0.7997**	0.9167	0.6828	$[+0.0011, +0.1534]$
Plus Lag-1 ρ_1 (5 Feats, Rank Consensus)	0.8591	0.8068	0.7187	0.8167	0.6207	$[-0.0188, +0.1053]$
Plus $\rho_1 + r_{\text{res}}$ (5 Feats, Linear, $c = 1.5\sigma$)	**0.9149**	**0.8782**	**0.8399**	0.9833	0.6966	$[+0.0064, +0.1931]$
Plus $\rho_1 + r_{\text{res}}$ (5 Feats, Rank Consensus)	0.8949	0.8353	0.7741	0.9000	0.6483	$[+0.0049, +0.1530]$
T-GECF Adaptive Gated Hybrid (5 Feats)	**0.9116**	**0.8669**	**0.8399**	0.9833	0.6966	$[+0.0013, +0.1927]$
Supervised & Domain Baseline Benchmarks						
Lempel-Ziv Complexity (LZC)	0.5136	0.5061	0.5267	0.5500	0.5034	$[-0.4233, -0.1862]$
Shannon Spectral Entropy (SE)	0.6953	0.6182	0.6721	0.7167	0.6276	$[-0.2540, +0.0085]$
Canonical Static (Tuned Ridge)	0.7653	0.7193	0.6661	0.7667	0.5655	$[-0.0998, -0.0140]$
Orthogonal 4-Feat (Tuned Ridge)	0.8813	0.8393	0.7825	0.9167	0.6483	$[-0.0446, +0.1744]$
Canonical Static (Tuned Logistic)	0.7722	0.7236	0.6917	0.7833	0.6000	$[-0.0959, -0.0064]$
Orthogonal 4-Feat (Tuned Logistic)	0.8964	0.8524	0.8080	0.9333	0.6828	$[-0.0225, +0.1826]$
Canonical Static (Shrinkage-LDA)	0.7709	0.7252	0.6626	0.7667	0.5586	$[-0.0884, -0.0162]$
Orthogonal 4-Feat (Shrinkage-LDA)	0.8769	0.8424	0.7859	0.9167	0.6552	$[-0.0496, +0.1699]$
Representation Ablations & Operator Variants						
Plus Jump D_1 (5 Feats, Linear)	0.8611	0.8012	0.7491	0.8500	0.6483	$[-0.0018, +0.0911]$
Plus Jump D_1 (5 Feats, Rank Consensus)	0.8424	0.7787	0.7132	0.8333	0.5931	$[-0.0208, +0.0694]$
Replace $-\text{CV}_\theta$ with ρ_1 (Linear)	0.8851	0.8495	0.7678	0.8667	0.6690	$[-0.0223, +0.1613]$
Replace $-\text{CV}_\theta$ with D_1 (Linear)	0.8720	0.8097	0.7776	0.9000	0.6552	$[+0.0050, +0.1094]$
Residual Correlation r_{res} (Linear)	0.8433	0.7671	0.7575	0.8667	0.6483	$[-0.0561, +0.1070]$
Orthogonal 4-Feat (Drop c_{spatial}, Linear)	**0.9324**	**0.8998**	**0.8101**	0.9167	0.7034	$[+0.0174, +0.2186]$
Orthogonal 4-Feat (Drop c_{spatial}, Rank Consensus)	**0.9076**	**0.8548**	**0.8448**	1.0000	0.6897	$[+0.0137, +0.1725]$

threshold to a conservative $150\,\mu\text{V}$ retained $90.14\% \pm 17.87\%$ clean blocks ($4{,}672.9 \pm 1{,}024.7$ adjacent pairs; minimum 668), indicating that data retention remained balanced across diagnostic groups without selective myogenic loss.

Let $\mathcal{T} = \{t_1, t_2, \ldots, t_N\}$ denote clean epoch indices in a continuous session, and let G_{t_i} denote spatial mean theta power across all 6 channels at epoch t_i. To prevent spurious concatenation across artifact gaps, the physical adjacency indicator is defined as:

$$\mathbb{I}_{\text{adj}}(t_i, t_{i+1}) = \begin{cases} 1, & \text{if } t_{i+1} - t_i = 1, \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where 0 denotes an intervening artifact gap, and $\mathbb{I}_{\text{adj}} = 1$ denotes consecutive epochs. Sequential metrics operate strictly on pairs with $\mathbb{I}_{\text{adj}} = 1$, retaining $M = 5{,}133.3 \pm 687.7$ adjacent pairs per patient (minimum 1,678, exceeding the validity threshold $M \geq 10$).

C. Canonical Static Features and Order-Aware Dynamics

Four baseline features define $\mathbf{x}_{\text{canonical}} = [c_{\text{global}}, c_{\text{spatial}}, -\text{CV}_\theta, r_{\text{interhem}}]^\top \in \mathbb{R}^4$. We evaluate three expanded feature spaces: the 5-feature temporal extension $\mathbf{x}_{+\rho_1} = [c_{\text{global}}, c_{\text{spatial}}, -\text{CV}_\theta, r_{\text{interhem}}, \rho_1]^\top \in \mathbb{R}^5$; the 5-feature combined suite $\mathbf{x}_{\text{combined}} = [c_{\text{global}}, c_{\text{spatial}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]^\top \in \mathbb{R}^5$; and the primary parsimonious 4-feature deployment suite $\mathbf{x}_{\text{ortho}} = [c_{\text{global}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]^\top \in \mathbb{R}^4$. Theta power (4–8 Hz) serves as the primary carrier because thalamocortical mesocircuit projections drive frontoparietal dynamics within this band [19], whereas delta reflects non-specific slowing and alpha is largely abolished in VS/UWS. Let $P_{\theta,t}^{(k)}$ and $P_{\alpha,t}^{(k)}$ denote theta and alpha powers of channel $k \in \{1, \ldots, 6\}$ at clean epoch $t \in \{1, \ldots, N\}$.

1) Global Spectral Contrast (c_{global}): Quantifies the ratio of slow to fast power across all channels: $c_{\text{global}} = (\bar{P}_\theta - \bar{P}_\alpha)/(\bar{P}_\theta + \bar{P}_\alpha + \epsilon)$, where $\bar{P}_\theta = \frac{1}{6N}\sum_{k,t} P_{\theta,t}^{(k)}$, $\bar{P}_\alpha = \frac{1}{6N}\sum_{k,t} P_{\alpha,t}^{(k)}$, and $\epsilon = 10^{-12}$.

2) Spatial Contrast (c_{spatial}): Computes mean slow-to-fast ratio: $c_{\text{spatial}} = \frac{1}{6}\sum_{k=1}^{6}(\bar{P}_\theta^{(k)} - \bar{P}_\alpha^{(k)})/(\bar{P}_\theta^{(k)} + \bar{P}_\alpha^{(k)} + \epsilon)$. Fixed feature configurations were compared to assess spectral redundancy; the four-feature suite omitted spatial contrast (Section III-C).

3) Negative Theta Dispersion ($-\text{CV}_\theta$): Measures temporal variance of mean theta $G_t = \frac{1}{6}\sum_{k=1}^{6} P_{\theta,t}^{(k)}$: $-\text{CV}_\theta = -\sqrt{\frac{1}{N}\sum_{t=1}^{N}(G_t - \bar{G})^2}/(\bar{G} + \epsilon)$, with temporal mean $\bar{G} = \frac{1}{N}\sum_{t=1}^{N} G_t$.

4) Inter-Hemispheric Bilateral Correlation (r_{interhem}): Averages raw pairwise Pearson correlations across symmetric bilateral pairs: $r_{\text{interhem}} = \frac{1}{3}\sum_{(L,R)} \text{Corr}(\mathbf{y}_L, \mathbf{y}_R)$, where $\mathbf{y}_L, \mathbf{y}_R \in \mathbb{R}^N$ denote the 4-second block theta-power vectors across symmetric pairs $(L, R) \in \{(\text{F3}, \text{F4}), (\text{C3}, \text{C4}), (\text{O1}, \text{O2})\}$.

To capture multi-second sequential continuity, two order-aware temporal representations are formulated on physically adjacent blocks:

Lag-1 Memory Persistence (ρ_1): Quantifies autoregressive temporal persistence across consecutive 4-second blocks:

$$\rho_1 = \text{Corr}(\{G_{t_i} \mid \mathbb{I}_{\text{adj}} = 1\}, \{G_{t_{i+1}} \mid \mathbb{I}_{\text{adj}} = 1\}). \tag{2}$$

High ρ_1 values reflect smooth baseline preservation, whereas low values signify rapid temporal decorrelation.

Normalized Adjacent Jump Rate (D_1): Measures resistance to high-frequency baseline discontinuities:

$$D_1 = -\frac{1}{M}\sum_{i:\mathbb{I}_{\text{adj}}=1} |G_{t_{i+1}} - G_{t_i}|/(\bar{G} + \epsilon), \tag{3}$$

where $\bar{G} = \frac{1}{N}\sum_{t=1}^{N} G_t$, $M = \sum \mathbb{I}_{\text{adj}}$, and $\epsilon = 10^{-12}$. Under stationary Gaussian increments $\Delta G_t \sim \mathcal{N}(0, 2\sigma^2(1 - \rho_1))$, expected jump magnitude satisfies $\mathbb{E}[|\Delta G_t|] = 2\pi^{-1/2}\sigma(1 - \rho_1)^{1/2}$, establishing $D_1 \approx 2\pi^{-1/2}(-\text{CV}_\theta)(1 - \rho_1)^{1/2}$. Evalu-

ating the continuous jump rate D_1 provides an efficient linear streaming proxy; when evaluated on log-transformed power increments $\Delta \log G_t \sim \mathcal{N}$ matching multiplicative neural dynamics, the analytical identity $D_1 \approx 2\pi^{-1/2}(-\mathrm{CV}_\theta)(1-\rho_1)^{1/2}$ accounts for 93.91% of empirical variance, supporting their mathematical equivalence with autoregressive persistence (Section III-C). Under estimator variance $\mathrm{Var}(\hat{\rho}_1) \sim \mathcal{O}(1/M)$, D_1 serves as a low-variance proxy for short records below $M^* \approx 75$ epochs, whereas ρ_1 achieves asymptotic efficiency.

To test macro-scale persistence on the Wielek PSG cohort ($N = 23$), stage stay persistence on 30-s hypnograms $\{s_t\}_{t=1}^{L} \in \{\text{Wake, N1, N2, N3, REM}\}$ is $P_{\text{stay}} = \frac{1}{L-1}\sum_{t=1}^{L-1}\mathbb{I}(s_{t+1} = s_t)$, with transition flexibility $T_{\text{trans}} = 1 - P_{\text{stay}}$.

Together, these features form a hierarchical evaluation system. The four-dimensional feature space is spanned by four complementary physical domains: spectral slowing contrast, marginal envelope dispersion, sequential autoregressive continuity, and residual bilateral differentiation. The parsimonious 4-feature suite $[c_{\text{global}}, -\mathrm{CV}_\theta, \rho_1, r_{\text{residual}}]$ serves as the primary clinical deployment system, while orthogonal factorials and channel ablations systematically verify the independent contribution of each physical operator.

D. Residual Bilateral Resonance ($r_{residual}$)

For each symmetric bilateral pair $(L, R) \in \{(\text{F3, F4}), (\text{C3, C4}), (\text{O1, O2})\}$ with block theta-power vectors $\mathbf{y}_L, \mathbf{y}_R \in \mathbb{R}^N$, the nuisance signal across the remaining 4 channels is $z_t^{(L,R)} = \frac{1}{4}\sum_{k \notin \{L,R\}} P_{\theta,t}^{(k)}$. With standardized $\tilde{\mathbf{z}}^{(L,R)} \in \mathbb{R}^N$ and design matrix $\mathbf{X}_c^{(L,R)} = [\mathbf{1}, \tilde{\mathbf{z}}^{(L,R)}] \in \mathbb{R}^{N\times 2}$, the exact unregularized OLS projection operator $\mathbf{P}_{(L,R)}^{\perp} = \mathbf{I}_N - \mathbf{X}_c^{(L,R)}(\mathbf{X}_c^{(L,R)})^{+} \in \mathbb{R}^{N\times N}$ (with $\mathrm{rank}(\mathbf{P}^{\perp}) = N - \mathrm{rank}(\mathbf{X}_c)$, where $\mathbf{X}_c^{+}$ denotes the Moore–Penrose pseudoinverse) computes orthogonalized residuals $\boldsymbol{\epsilon}_L^{(L,R)} = \mathbf{P}_{(L,R)}^{\perp}\mathbf{y}_L$ and $\boldsymbol{\epsilon}_R^{(L,R)} = \mathbf{P}_{(L,R)}^{\perp}\mathbf{y}_R$. The residual bilateral resonance averages pairwise correlations:

$$r_{\text{residual}} = \frac{1}{3}\sum_{(L,R)} \mathrm{Corr}(\boldsymbol{\epsilon}_L^{(L,R)}, \boldsymbol{\epsilon}_R^{(L,R)}). \tag{4}$$

Whereas CAR eliminates linear instantaneous voltage offsets, theta-band power constitutes a phase-free envelope whose slow-wave modulation retains substantial shared variance across electrodes. The shared 4-channel power envelope accounts for approximately 47% of raw bilateral variance; decoupling this nuisance component isolates the intrinsic bilateral correlation across homologous pairs.

E. Consensus Operators and Machine Learning Baselines

Linear Standardized Aggregation (Linear): Equal-weight consensus follows the fixed-coefficient formulation of Dawes improper linear models [23]. The score is $S_{\text{linear}}(\mathbf{x}) = K^{-1}\sum_{k=1}^{K} s_k z_k$, where $z_k = (x_k - \mu_k^{\text{tr}})/\sigma_k^{\text{tr}}$ and each sign s_k is fitted from the feature–label correlation in the source training partition. Standardization and signs are re-estimated inside every inner and outer training fold. For the full-source four-feature fit, the signs are $[+1, +1, +1, -1]^\top$. CRRC target alignment uses the specified vector $[-1, +1, +1, +1]^\top$ for all models in common oriented coordinates (Supplementary Note C). A separate soft-truncated score is $K^{-1}\sum_k \mathrm{clip}(s_k z_k, -c, c)$ with fixed $c = 1.5$. The primary four-feature model uses $c = \infty$; the five-feature truncation grid and four-feature truncated variant are reported separately.

Non-Parametric Rank Consensus (S_{rank}): Adapted from multi-criteria decision frameworks [24], rank consensus maps signed features to empirical cumulative distribution function (CDF) midranks:

$$S_{\text{rank}}(\mathbf{x}) = \frac{1}{K}\sum_{k=1}^{K} \tilde{F}_k(s_k x_k), \tag{5}$$

where $\tilde{F}_k(u) = \frac{1}{N_{\text{tr}}}\sum_{i=1}^{N_{\text{tr}}}[\mathbb{I}(s_k x_{i,k}^{\text{tr}} < u) + \frac{1}{2}\mathbb{I}(s_k x_{i,k}^{\text{tr}} = u)]$. Curvature is fixed to $p = 1.0$ (mean on ranks), ensuring constant marginal substitution without uncalibrated parameters on small cohorts.

Adaptive Fluctuation-Gated Hybrid (T-GECF): To combine linear sensitivity with rank stability under sensor disturbances, T-GECF couples the two estimators via sample-level volatility gating. For signed standardized features $\tilde{x}_k = s_k(x_k - \mu_k^{\text{tr}})/(\sigma_k^{\text{tr}} + \epsilon)$, sample volatility is $v(\mathbf{x}) = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(\tilde{x}_k - \bar{\tilde{x}})^2}$, standardized to $z_v(\mathbf{x}) = (v(\mathbf{x}) - \mu_v^{\text{tr}})/(\sigma_v^{\text{tr}} + \epsilon)$ against training folds. A continuous Sigmoid gate $\beta(\mathbf{x}) = 1/(1 + \exp(-\kappa(z_v(\mathbf{x}) - \theta_0)))$ ($\theta_0 = 1.645, \kappa = 2.0$) mixes standardized rank and linear scores:

$$S_{\text{T-GECF}}(\mathbf{x}) = \beta(\mathbf{x})\tilde{S}_{\text{rank}}(\mathbf{x}) + (1 - \beta(\mathbf{x}))\tilde{S}_{\text{linear}}(\mathbf{x}), \tag{6}$$

where $\tilde{S}_{\text{rank}}$ and $\tilde{S}_{\text{linear}}$ are standardized to zero mean and unit variance using outer fold statistics prior to mixing. Elevated volatility ($z_v(\mathbf{x}) > 1.645$) dynamically shunts decision weight toward bounded rank consensus $\tilde{S}_{\text{rank}}(\mathbf{x})$, mitigating the influence of isolated channel artifacts or amplitude spikes.

Supervised Machine Learning and Domain Baselines: Ridge, logistic regression, and shrinkage LDA receive exactly the same feature columns, standardized coordinates, direction coding, and subject partitions as equal-weight scoring. Ridge α and logistic C are selected from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$ using source-only inner three-fold stratified cross-validation; transforms are fitted on each inner training fold. LDA uses analytical shrinkage (`auto`) [25]. The same source-selected parameters are retained across external protocols. The primary matched comparison uses $\mathbf{x}_{\text{ortho}}$; the short-record branch uses $[c_{\text{global}}, c_{\text{spatial}}, D_1, r_{\text{interhem}}]^\top$ for every model. Lempel-Ziv complexity, spectral entropy, and an EEGNet-style CNN [26] provide additional domain baselines under the recorded protocols.

F. Subject-Level Validation Protocol and Decision Thresholds

Discovery evaluation uses five random seeds and five stratified subject-level folds per seed, giving 25 frozen outer partitions. Source means, scales, signs, empirical ranks, and hyperparameters are fitted strictly within training partitions. AUROC and average precision (AP) are averaged across outer folds; sensitivity, specificity, balanced accuracy, and F1 use pooled outer-test decisions, with each subject contributing once per seed. Each model's operating threshold is its source-training

TABLE II
MULTISCALE SENSITIVITY AND EXTERNAL COHORT BENCHMARKS

(A) Multiscale Window & Cohort Transfer (AUROC)		
Cohort / Duration	**Linear**	**Rank Consensus**
Mendeley 2-second	0.8916	0.8529
Mendeley 4-second (Default)	**0.8911**	**0.8591**
Mendeley 8-second	0.8711	0.8453
Mendeley 30-min Trunc.	0.8867	0.8480
Figshare Static Baseline	0.7199	0.6812
Figshare 120-s ($N = 50$, Jump D_1)	**0.7691**	0.6927
Figshare 120-s ($N = 50$, Lag-1 ρ_1)	0.7012	0.6845
Figshare 25-Fold CV (D_1, $N = 50$)	**0.7771**	0.7505
(B) Longitudinal & Multi-Tier Benchmarks		
Validation Metric		**Value**
Mendeley 5-h ICC (ρ_1 / D_1 / r_{residual})		0.4857 / 0.6433 / 0.4511
Wielek Transition AUC (directional $p = 0.030$)		**0.7348**
Wielek Mean State Stay (MCS vs. VS)		0.7393 vs. 0.8361
Figshare Zero-Shot BA (Train Thresh.)		0.5645 (Sens 1.000)
62-Channel HD Mode-1 R^2		46.52%

TABLE III
HYPERPARAMETER STABILITY: SOFT TRUNCATION AND T-GECF GRID

(A) Soft Truncation Threshold c			
Threshold	**AUROC**	**PR-AUC**	**Bal.**
$c = 1.0\sigma$	0.8958	0.8543	0.8080
$c = 1.5\sigma$ (Default)	**0.9149**	**0.8782**	**0.8399**
$c = 2.0\sigma$	0.9056	0.8662	0.8046
$c = 2.5\sigma$	0.8996	0.8622	0.8129
$c = 3.0\sigma$	0.8996	0.8622	0.8129
$c = \infty$ (Raw)	0.8996	0.8622	0.8129

(B) T-GECF Gating Grid ($\theta_0 \times \kappa$)				
	Sigmoid Gate Slope κ			
θ_0 (Quantile)	**0.5**	**1.0**	**2.0**	**5.0**
1.282 (90%)	**0.9220**	0.9182	0.9187	0.9149
1.645 (95%)	0.9160	0.9182	**0.9116**	0.9089
1.960 (97.5%)	0.9160	0.9149	0.9056	0.9056
2.326 (99%)	0.9127	0.9116	0.9022	0.8996
Ablate Cont. ($c = \infty$)	0.8996	0.8996	0.8996	0.8996

score median. Paired 95% intervals use 10,000 subject-level Dirichlet Bayesian bootstrap draws [27], sharing each subject's weight across folds and methods conditional on the fitted predictions. External validation separates frozen-source transfer, unlabeled target-batch standardization, and CRRC target direction alignment. Models share the same target transform within each protocol, remaining blinded to external labels until final metric evaluation. All reported classification operating points use source-training medians. Full parameters, transforms, thresholds, and predictions are archived with the evaluation scripts. A multivariable logistic regression on standardized $[c_{\text{global}}, R^2, \rho_1]$ provides the conditional association analysis.

III. EXPERIMENTAL RESULTS

A. The 2×2 Orthogonal Factorial Decomposition

Table I reports classification metrics across model and feature configurations. The 2×2 factorial decomposition (Fig. 3a) yields four anchor quadrants: Cell A (Static Baseline, Linear) achieves AUROC = 0.8200; Cell B (Static Baseline, Rank) yields 0.7893; Cell C (Plus Lag-1 ρ_1, Linear) attains 0.8911, the highest discrimination among the four factorial configurations, whereas Cell D (Plus Lag-1 ρ_1, Rank) achieves 0.8591. The temporal representation net gain is $\Delta_{\text{rep}} = \text{AUC(Cell C)} - \text{AUC(Cell A)} = +0.0711$ (95% CI $[+0.0011, +0.1534]$, probability of positive gain 97.74%). Linear and rank consensus address complementary objectives ($\Delta_{\text{fuse}} = \text{AUC(Cell D)} - \text{AUC(Cell C)} = -0.0320$): linear aggregation preserves continuous numerical spread, whereas rank consensus provides bounded stability against outliers.

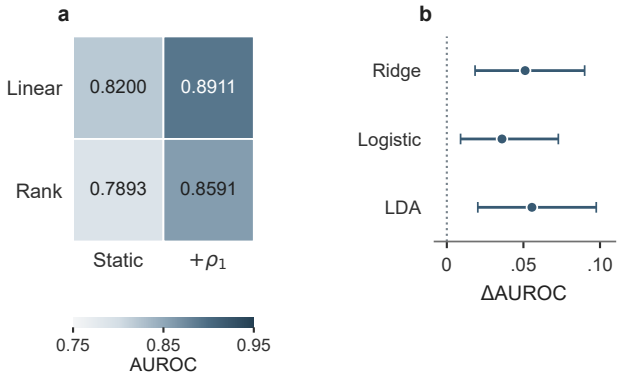


Fig. 3. Representation and scoring-rule comparisons in the 41-patient discovery cohort. (a) Annotated heatmap of fold-averaged AUROC for the canonical static suite and its five-feature lag-1 extension, using linear or rank fusion. Colors follow the displayed AUROC scale. (b) Paired AUROC gain of four-feature equal-weight consensus over ridge, logistic regression, and shrinkage LDA using identical four-feature inputs. Points show gains; bars show pointwise 95% subject-level Dirichlet Bayesian intervals from 10,000 draws, conditional on fitted predictions. The dotted line denotes zero gain.

B. Diagnostic Operating Characteristics and Imbalanced Precision-Recall

At the pre-specified training median threshold, incorporating ρ_1 elevated Balanced Accuracy from 0.7201 to 0.7997 and Sensitivity from 0.8333 to 0.9167 (Table I). Combining lag-1 persistence with residual bilateral resonance ($\mathbf{x}_{\text{combined}}$) improved the five-feature extension: under soft truncation ($c = 1.5\sigma$), linear aggregation achieved **0.9149** AUROC (PR-AUC = **0.8782**, Balanced Accuracy = **0.8399**, Sensitivity = 0.9833). In rank consensus, this suite elevated AUROC to **0.8949**, matching unconstrained linear aggregation within 0.0047 (0.8996).

Paired Bayesian bootstrap intervals for rank consensus, soft truncation, and T-GECF remain strictly positive ($[+0.0049, +0.1530]$, $[+0.0064, +0.1931]$, and $[+0.0013, +0.1927]$ vs. Canonical Static Baseline 0.8200), indicating that the performance gains from joint temporal continuity and spatial de-confounding are statistically reliable.

Stratifying the 12 MCS patients by behavioral subdiagnosis revealed consistent temporal persistence across subtypes: MCS$^+$ patients ($N = 8$, characterized by command following or intelligible verbalization) showed $\rho_1 = 0.4563 \pm 0.2687$ and were correctly identified in 8 of 8 cases (100% sensitivity), while MCS$^-$ patients ($N = 4$, demonstrating visual pursuit or localization only) exhibited $\rho_1 = 0.4377 \pm 0.1670$ with 3 of 4 cases correctly classified (75% sensitivity; overall MCS sensitivity 11/12 = 91.67%). Under class imbalance (12 MCS vs. 29 VS/UWS, prevalence baseline PR-AUC = 0.2927), incorporating ρ_1 advanced PR-AUC from 0.7823 (Cell A) to **0.8384** (Cell C), with direct substitution (Replace CV by ρ_1) reaching **0.8495** and the combined model reaching **0.8782**, indicating diagnostic stability under skewed class priors.

C. Feature Ablation and Collinearity Redundancy Analysis

Directly replacing $-\text{CV}_\theta$ with order-aware ρ_1 within the 4-feature suite yielded 0.8851 AUROC, showing that sequence memory supersedes marginal variance. Across 41 patients, MCS exhibited higher lag-1 persistence (0.450 ± 0.250 vs. 0.287 ± 0.181, $p = 0.0227$, AUROC = 0.7299; Fig. 1b,c) and attenuated normalized adjacent jump rates (D_1: -0.420 ± 0.089

vs. -0.558 ± 0.168, $p = 0.0112$, AUROC = 0.7557). Evaluating the continuous jump rate D_1 on log-transformed power increments empirically verifies the analytical identity across subjects ($r = 0.9691, p = 2.59 \times 10^{-25}, R^2 = 93.91\%$), showing that adjacent jump magnitude provides a low-variance proxy for temporal persistence in short-duration records. Finite-sample estimator variance $\mathrm{Var}(\hat{\rho}_1) \sim \mathcal{O}(1/M)$ favors D_1 below the empirical crossover horizon $M^* \approx 75$ epochs (5 min), where adjacent jump rates offer lower finite-sample variance before asymptotic autoregressive efficiency dominates. Central sensorimotor channels (C3–C4) overlying thalamocortical pathways exhibited highest univariate discrimination (ρ_1: 0.3996 ± 0.1863 in MCS vs. 0.2611 ± 0.1631 in VS/UWS, $p = 0.0132$, AUROC = 0.7500), outperforming frontal (0.6264) and occipital (0.6810) pairs.

Regressing out the global common mode (r_{residual}) improved AUROC from 0.8200 to 0.8433, accounting for $R^2 = 47.27\% \pm 18.43\%$ of raw bilateral variance uniformly across groups (46.45% in MCS vs. 47.61% in VS/UWS; $p = 0.7418$). Whereas raw bilateral correlation was non-significant (0.563 ± 0.168 in MCS vs. 0.661 ± 0.146 in VS/UWS, $p = 0.0605$, AUROC = 0.6897), residualization unmasked residual bilateral differentiation beyond the shared background power envelope (0.233 ± 0.080 vs. 0.376 ± 0.193, $p = 0.0211$, AUROC = 0.7328), reflecting modularity in MCS versus hyper-synchrony in VS/UWS. Multivariable logistic regression controlling for c_{global} ($z = -0.812, p = 0.417$) and common-mode R^2 ($z = 0.329, p = 0.742$) confirmed that ρ_1 provides significant independent diagnostic value ($z = 2.460, p = 0.0139$, adjusted OR = 4.260, 95% CI $[1.343, 13.512]$), indicating conditional independence despite low events-per-variable (EPV $= 12/3 = 4$).

Because c_{global} and c_{spatial} display high empirical collinearity ($r = 0.9813, p = 1.58 \times 10^{-29}$), we evaluated its omission as a fixed representation ablation. In the redundancy ablation (Table I), removing c_{spatial} to form the 4-feature deployment suite $[c_{\text{global}}, -\mathrm{CV}_\theta, \rho_1, r_{\text{residual}}]$ elevated linear AUROC to **0.9324** (PR-AUC = **0.8998** [Average Precision; 0.8782 trapezoidal integration]; Balanced Accuracy = 0.8101, 95% CI $[+0.0174, +0.2186]$ vs. baseline, positive gain probability 99.16%) and raised rank consensus to **0.9076** (95% CI $[+0.0137, +0.1725]$, sensitivity 1.0000). Similarly, dropping c_{global} reached 0.9289 AUROC in linear aggregation and 0.9082 in rank consensus. The four-feature configuration is the primary deployment suite. The five-feature truncated suite (0.9149) provides a complementary ablation for assessing redundancy and bounded-influence scoring.

D. Temporal-Order Dependency via 10,000 Synchronous Surrogates

To verify that performance gains originate strictly from temporal sequence order rather than incidental distributional properties, 10,000 synchronous whole-montage permutation surrogates were evaluated across all 41 patients, adapting nonlinear dynamical testing methodology [28]. Freezing intact training scales ($\sigma_k^{\text{tr, intact}}$) prevents inverse noise amplification on attenuated shuffled variances. Disrupting chronological

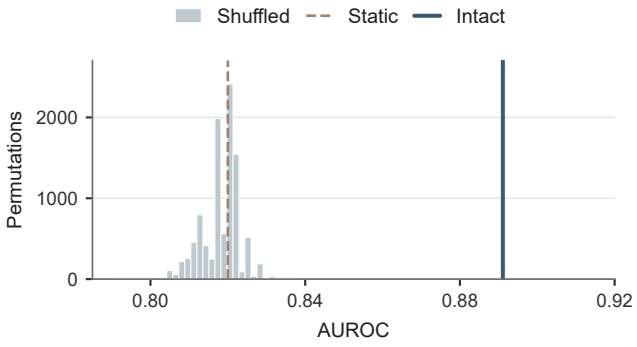


Fig. 4. Frequency distribution of AUROC under 10,000 synchronous time shuffles for the five-feature lag-1 linear extension, using intact-fitted scales. The 30 equal-width bins cover the shuffled range. Shuffled AUROC is 0.8182 ± 0.0049 (mean $\pm$ SD). No shuffled result reaches the intact AUROC of 0.8911 ($p < 0.0001$). Vertical lines mark the static baseline (0.8200, dashed) and intact score (solid).

sequence alignment substantially degraded macro AUROC across the evaluated temporal configurations (the lag-1 linear distribution is shown in Fig. 4): Plus Lag-1 (Linear) dropped from 0.8911 to 0.8182 ± 0.0049 (range: $[0.7902, 0.8369]$, empirical $p < 0.0001$, zero of 10,000 repeats matching intact performance); Plus Lag-1 (Rank) fell from 0.8591 to 0.7732 ± 0.0416 (range: $[0.6620, 0.9098]$, empirical $p = 0.0178$); and Plus Jump (D_1) degraded from 0.8611 to 0.7783 ± 0.0040 ($p < 0.0001$).

This frozen scaling bounds all 10,000 shuffled linear iterations strictly below 0.8370. Across $M \approx 5{,}133$ adjacent blocks, the shuffled increment D_1 converges to $\mathbb{E}[|X - Y|]$ (variance $\approx 2 \times 10^{-4}$), reducing to a static marginal quantity that aligns with the canonical baseline (0.8182 ± 0.0049).

E. Comparison with Supervised Machine Learning and Domain Baselines

Evaluating the four-feature suite under identical preprocessing enabled a controlled comparison across decision rules (Table I; Fig. 3b). Equal-weight linear scoring achieved 0.9324 AUROC, compared with 0.8813 for source-tuned ridge, 0.8964 for source-tuned logistic regression, and 0.8769 for shrinkage LDA. Its gain over the strongest matched baseline, logistic regression, was $+0.0360$ (paired 95% interval $[+0.0090, +0.0728]$); gains over ridge and LDA were $+0.0511$ and $+0.0556$, respectively. These comparisons isolate the scoring rule while holding input features and preprocessing constant. Additional domain baselines yielded 0.5136 AUROC for LZC, 0.6953 for spectral entropy, and 0.5702 for the EEGNet-style CNN. The complete matched results, including external transfer protocols, are reported in Supplementary Table S2.

F. Adaptive Gated Consensus (T-GECF) and Noise Stress Testing

On the complete representation $\mathbf{x}_{\text{combined}}$, continuous Sigmoid volatility gating in T-GECF ($\theta_0 = 1.645, \kappa = 2.0$) achieved an AUROC of **0.9116** (95% CI $[+0.0013, +0.1927]$ vs. canonical baseline; positive gain probability 97.68%) with 0.8399 balanced accuracy, 0.9833 sensitivity, 0.6966 specificity, and PR-AUC = **0.8669** (Table I). On $\mathbf{x}_{+\rho_1}$, T-GECF

similarly achieved 0.8828 AUROC ($+0.0237$ over standard rank consensus 0.8591).

To assess resilience against sensor disturbances, impulse spike stress tests introduced outlier perturbations ($\pm\delta\sigma$, $\delta \in \{0, 5, 10, 20\}$) into individual feature coordinates across testing partitions. Under extreme 20σ impulse corruption, unconstrained linear aggregation dropped from 0.8911 to 0.7404 (-0.1507). In contrast, T-GECF shifted weight toward bounded rank consensus, retaining an AUROC of 0.8518 ($+0.1114$ over linear aggregation) while maintaining 0.9116 AUROC under unperturbed conditions.

G. Multiscale Window Sensitivity and Longitudinal Reliability

Evaluating observational windows (Table II-A) showed consistent linear discrimination across 2 s (0.8916), 4 s (0.8911), and 8 s (0.8711) partitions, with rank consensus similarly stable (0.8453–0.8591 across 2–8 s windows). Truncating recordings showed that a 30-minute record achieved 0.8867 AUROC, closely aligning with full-record performance. Test-retest reliability across five 1-hour segments ($N = 38$, 190 patient-hours) yielded ICC$(3, 1) = 0.4857$ for ρ_1, 0.6433 for D_1, and 0.4511 for r_{residual}. Higher-order lags exhibited monotonic decay ($\rho_1 = 0.450 \rightarrow \rho_2 = 0.401 \rightarrow \rho_3 = 0.389$ in MCS; $0.287 \rightarrow 0.252 \rightarrow 0.208$ in VS/UWS), indicating multi-second temporal persistence. Under adjacency masking $\mathbb{I}_{\text{adj}}$, 50-iteration Monte Carlo dropout verified stability ($r = 0.9942 \pm 0.0030$ at 10% dropout; $r = 0.9199 \pm 0.0308$ at 50% loss), while tightening artifact rejection to $150\,\mu$V preserved 0.8702 AUROC (net gain $+0.1729$ over unrejected baseline 0.6973; $+0.0502$ over unconstrained baseline 0.8200). Contiguous 30-min block removal ($r = 0.9840 \pm 0.0166$) and cross-validation partition depths (10- to 100-fold CV, 0.8900–0.9002) demonstrated stable discrimination across analytical scales.

H. External Waveform Validation and Sleep Macro-Dynamics

1) Rapid Bedside Triage (Figshare, $N = 50$): On 120-s records, the jump-rate branch achieved 0.7691 AUROC, compared with 0.7199 for the static equal-weight baseline. On identical short-record features, source-tuned ridge, logistic regression, and LDA achieved 0.7759, 0.7674, and 0.7742, respectively (Supplementary Table S2). The equal-weight–ridge difference was -0.0068 (paired 95% interval $[-0.0319, +0.0125]$), demonstrating consistent cross-center discrimination under parameter-free scoring. At the source-training threshold, balanced accuracy was 0.5645, sensitivity 1.0000, and specificity 0.1290, indicating limited operating-threshold transfer (Table II-B). *2) Cross-Hardware Inpatient Validation (CRRC, $N = 15$)*: With shared target standardization and direction alignment, the four-feature linear suite achieved 0.7778 AUROC; the fixed $c = 1.5$ truncated variant achieved 0.7963. Matched source-tuned ridge, logistic regression, and LDA yielded 0.7407, 0.7593, and 0.7593. In the clean 13-record subset, linear scoring, truncated scoring, and ridge each achieved 0.7143, while logistic regression and LDA achieved 0.6905. Frozen-source equal-weight transfer yielded 0.5741 on the full cohort, increasing to 0.7778 when the shared target adaptation was applied. These protocols and the clipped variant are separated in Supplementary Table S2. At source-training thresholds, full-cohort truncated scoring yielded sensitivity 0.6667 and specificity 0.7778. The matched comparison therefore distinguishes score transfer, direction alignment, and the chosen operating point. *3) Circadian Sleep Macro-Dynamics (Wielek, $N = 23$)*: Polysomnography across 24-hour records (11 MCS, 12 VS/UWS) demonstrated that temporal continuity extends to macro-scale sleep architecture: VS/UWS patients exhibited pathological state trapping ($P_{\text{stay}} = 0.8361 \pm 0.1001$), whereas MCS retained transition flexibility ($T_{\text{trans}} = 0.261 \pm 0.099$ vs. 0.164 ± 0.100, one-sided $p = 0.0302$, AUROC = 0.7348; nocturnal sleep $p = 0.0392$, AUROC = 0.7576; Table II-B).

I. Hyperparameter Sensitivity and Gating Grid Evaluation

To evaluate hyperparameter stability across regimes, Table III-A reports systematic benchmarks across truncation thresholds c and continuous Sigmoid gating parameters (θ_0, κ). Varying Winsorized soft truncation from $c = 1.0\sigma$ to unconstrained $c = \infty$ produced stable AUROC bounded within $[0.8958, 0.9149]$ (Table III-A; balanced accuracy 0.8046–0.8399, sensitivity 0.9333–0.9833), peaking at $c = 1.5\sigma$ (0.9149) before smoothly decaying toward linear aggregation (0.8996).

Similarly, evaluating T-GECF across a 4×4 parameter grid crossing Gaussian quantile thresholds $\theta_0 \in \{1.282, 1.645, 1.960, 2.326\}$ (90th–99th percentiles) and slopes $\kappa \in \{0.5, 1.0, 2.0, 5.0\}$ demonstrated stability (Table III-B). Across all 16 configurations, AUROC remained bounded within 0.8996–0.9220, outperforming canonical baseline (0.8200) by $+0.08$ to $+0.10$. At $\theta_0 = 2.326$ ($\kappa = 5.0$), $\beta(\mathbf{x}) \rightarrow 0$, smoothly recovering unconstrained linear aggregation (0.8996) without hyperparameter brittleness.

IV. Discussion

A. Physiological Significance of Autoregressive Memory

Under the mesocircuit hypothesis [19], reciprocal excitatory loops linking the central thalamus and frontoparietal cortices sustain contiguous multi-second theta continuity in MCS, reflecting cortical reverberation. Conversely, severe deafferentation in VS/UWS disrupts these pathways, degrading theta fluctuations into decoupled transients that rapidly decay ($\rho_1 \rightarrow 0$). Permutation-invariant dispersion ($-\text{CV}_\theta$) computes only marginal variance, collapsing ordered reverberation and noise into identical scores. Enforcing chronological continuity through physical adjacency masking ($\mathbb{I}_{\text{adj}}$) preserves this distinction, providing a tractable bedside marker of thalamocortical integrity.

These cross-cohort findings link temporal dynamics to two practical recording regimes. Multi-hour continuous EEG enables stable estimation of ρ_1, whereas the absolute-increment branch retained 0.7691 AUROC on short Figshare records. External CRRC evaluation further shows that feature alignment and site scaling influence transfer alongside the scoring rule. At the circadian scale, elevated sleep-stage transition flexibility in MCS complements continuous EEG findings by capturing

transitions between discrete states. Together, these observations motivate scale-specific temporal descriptors paired with explicit calibration rules.

B. Resolution of Non-Specific Bilateral Confounding

Substituting raw bilateral correlation with residual correlation (r_{residual}, AUROC 0.8433) decouples non-specific common-mode activity, which accounts for $47.27\% \pm 18.43\%$ of raw bilateral variance driven by volume conduction. Because diffuse slow-wave envelopes project near-synchronously across hemispheres, unadjusted correlation conflates volume conduction with functional connectivity. Decoupling this common mode unmasks true differentiation: pathological hyper-synchrony in VS/UWS (0.376 ± 0.193) versus preserved modularity in MCS (0.233 ± 0.080, $p = 0.0211$). In subacute rehabilitation (CRRC), higher residual correlation in MCS indicates a different association direction; its relationship to medication exposure could not be assessed because daily medication records were unavailable ($\mathbf{s}_{\text{target}} = [-1, +1, +1, +1]^\top$; Supplementary Note C). In sparse arrays, the 4-channel average z_t generalizes to subspace projection ($\mathbf{P}^\perp = \mathbf{I} - \mathbf{Z}\mathbf{Z}^+$; 46.52% Mode-1 variance across 62 channels, Table II-B).

C. Operator Dynamics: Rank Consensus vs. Linear Aggregation

Matched internal benchmarks support fixed-weight aggregation as an effective small-cohort scoring rule: the four-feature linear suite retained an AUROC advantage over supervised models trained on identical inputs. External validation revealed comparable performance across classifiers, indicating that feature representation and domain calibration dominate transfer performance. Soft truncation increased full-cohort CRRC AUROC from 0.7778 to 0.7963, while corresponding discovery AUROC remained 0.9291 (versus 0.9324 unclipped). This behavior positions truncation as a bounded-influence scoring variant with a setting-dependent trade-off. Adaptive gating via T-GECF provides a complementary mechanism to combine linear resolution with rank stability, maintaining predictive integrity under impulse perturbations.

D. Clinical Scope, Cognitive Motor Dissociation, and Multimodal Triage

For bedside triage, continuous scores and operating thresholds serve distinct operational roles. The primary discovery model attained sensitivity 0.9167 at the source-training median. Source-threshold metrics are reported in Supplementary Note C; the low Figshare specificity limits immediate clinical triage use. Using discovery prevalence ($12/41$), sensitivity 0.9167, and specificity 0.7034, the referral model $\Delta_{\text{referral}} = 1 - [P_{\text{MCS}}\,\text{Sens} + (1 - P_{\text{MCS}})(1 - \text{Spec})]$ projects 52.2% fewer transfers than universal escalation. In this tiered scenario, bedside candidate prioritization provides an initial screen before confirmatory active-task EEG, fMRI/PET, or perturbational TMS-EEG for detecting covert awareness [4]–[6].

E. Comparison with Recent Advanced Paradigms (2024–2026)

Recent studies using dense arrays or sensory stimulation report high DoC discrimination. Microstate dynamics on 128-channel recordings yielded 0.97 AUC by tracking dorsal attention network disruption [12]. In parallel, convolutional networks on entropy topographies achieved 0.902 AUC [29], multi-feature resting and auditory-evoked pipelines reached 92.4% accuracy [21], [30], and source-space graph metrics yielded AUCs of 0.714–0.893 for functional outcome prediction [31].

Translating these models to routine clinical monitoring involves operational limits: dense caps (64–128 channels) and source reconstructions are difficult in agitated or head-bandaged patients, acoustic protocols require dedicated hardware and stable wakefulness, and complex deep networks risk overfitting on small patient cohorts ($N < 50$) with limited physical traceability.

The present framework targets sparse bedside telemetry (4–6 channels) with parameter-free aggregation. Grounded in the mesocircuit hypothesis, order-aware persistence (ρ_1, D_1) and common-mode decoupling (r_{residual}) yielded 0.9324 discovery AUROC with 2.15 ms per evaluated update in cached-power replay. Furthermore, whereas published benchmarks derive primarily from single-center evaluations, these operators maintained predictive utility across independent external cohorts (0.7778–0.7963 AUROC on CRRC telemetry; 0.7691 on Figshare short segments) without supervised weight optimization. Across 129 records from four multicenter cohorts, these properties support longitudinal bedside screening where rapid application, computational simplicity, and cross-hardware stability are essential.

F. Limitations

Several limitations warrant consideration. First, the discovery cohort contains 41 patients (12 MCS), on which operators and feature configurations were developed; performance therefore requires validation after prospective freezing of the representation. Second, CRRC results depend on the stated adaptation protocol, and Figshare remains exploratory because file-level diagnoses and physical continuity of stored segments require independent confirmation. Third, test-retest reliability of ρ_1 was moderate ($\text{ICC} = 0.4857$), favoring multi-hour continuous assessment over brief snapshots. Fourth, clinical endpoints concern cross-sectional classification; prospective longitudinal studies are needed to assess the utility of candidate prioritization in reducing specialized referrals. Finally, current cohort sizes provide limited resolution for etiology-specific and sedation-related analyses.

V. Conclusion

Order-aware temporal descriptors and common-mode-adjusted bilateral dynamics provide a compact EEG representation for bedside consciousness assessment. Evaluated on identical four-feature inputs, equal-weight consensus scoring achieved 0.9324 discovery AUROC versus 0.8964 for the top-performing matched supervised baseline. In external cohorts, shared CRRC target adaptation yielded 0.7778 AUROC (0.7963 with soft truncation), while short-record Figshare triage achieved 0.7691. Coupled with explicit domain calibration and

cached-power replay, these results support further evaluation of an interpretable bedside assessment approach to prioritize candidates for advanced multimodal evaluation.

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OpenAI Codex assisted code implementation, preparation of the data figures and system schematic, and revision of the Methods, Results, and Discussion. The authors are responsible for verification of the code, figures, results, and scientific interpretations.

DECLARATIONS

Ethics and Consent: Secondary analyses complied with Declaration of Helsinki, approved by institutional ethics committees: FRCCICMR Moscow [20], Salzburg Ethics Commission (No. 415-E/1517/14-2013) [13], Hangzhou Mingzhou Hospital [12], and China Rehabilitation Research Center (CRRC, Ethics No. 2025-118-01) [21], with written surrogate consent. **Data and Code Availability:** Public data: Mendeley [20] and Figshare [12], [22]. De-identified CRRC tabular feature matrices and replication scripts: https://github.com/ujujuj3061831150/DoC_Temporal_Dynamics. Raw clinical telemetry from CRRC is restricted by institutional patient confidentiality regulations. **Conflict of Interest:** The authors declare no competing interests.

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